

Encyclopedia of Plant Science - Photosynthetic Pathways

Photosynthesis sustains almost all higher life forms on Earth by fueling the base of ecological food webs and generating atmospheric oxygen. Plants utilize solar radiation to drive the reduction of carbon dioxide into organic starch and sugars.

C3, C4, and CAM pathways represent evolutionary adaptations in plant physiology to maximize carbon assimilation under varying moisture, temperature, and arid environmental stresses.

Stomata on the underside of plant leaves regulate transpiration and gas exchange, allowing carbon dioxide intake while minimizing water vapor loss during intense sunlight.